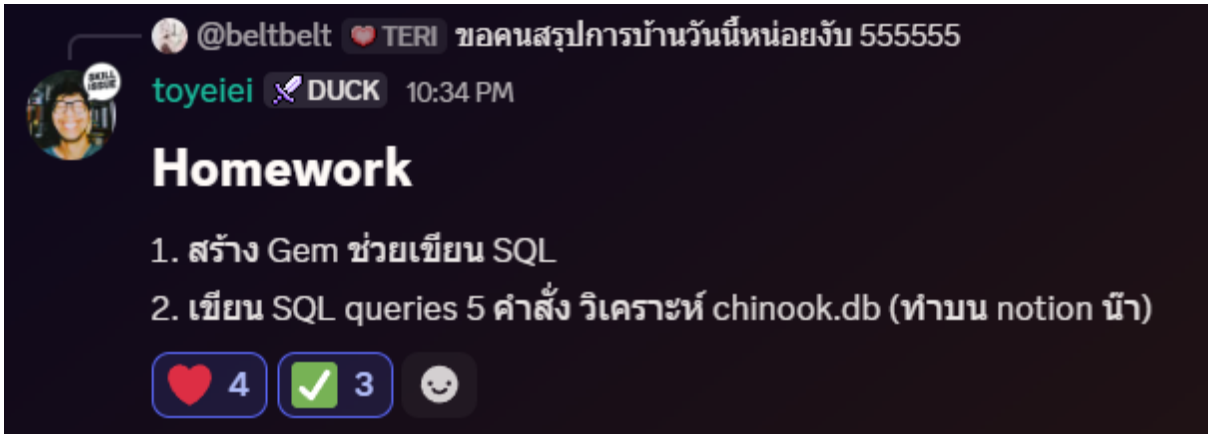
5 Scripts of SQL

Status	Done
Assign	Pasit Promchawee
Classes	Live 05 - SQL Database

Write any 5 scripts based on `chinook.db`.



QUERY #1: Find top 10 count of albums by artists
QUERY #2: Find report to manager of all employees
QUERY #3: Find big spender by country
QUERY #4: Find total file size in each albums and artist
QUERY #5: Create file size segment

QUERY #1: Find top 10 count of albums by artists

AS the header said, I need `artists` and `albums` table by joining them.

Then I use `COUNT()` as aggregation and group by artist category.

```sql
SELECT
        ar.Name,
    COUNT(*) AS count_albums
FROM artists ar
JOIN albums al ON ar.ArtistId = al.ArtistId
GROUP BY ar.Name
ORDER BY COUNT(*) DESC
LIMIT 10
```

Name	count_albums
Iron Maiden	21
Led Zeppelin	14
Deep Purple	11
U2	10
Metallica	10
Ozzy Osbourne	6
Pearl Jam	5
Various Artists	4
Van Halen	4
Lost	4

QUERY #2: Find report to manager of all employees

This one use “SELF JOIN” technique where I notice on ER diagram here:

And I have to be careful with join key, This one have to be `em.ReportsTo = emj.EmployeeId` . If I swapped, the data will swap between employee and manager which it’s wrong.

```
SELECT
        em.firstname || ' ' || em.lastname AS employee_name,
    em.Title AS employee_position,
    emj.firstname || ' ' || emj.lastname AS manager_name,
    emj.Title AS manager_position
FROM employees em
JOIN employees emj ON em.ReportsTo = emj.EmployeeId
```

employee_name	employee_position	manager_name	manager_position
Nancy Edwards	Sales Manager	Andrew Adams	General Manager
Jane Peacock	Sales Support Agent	Nancy Edwards	Sales Manager
Margaret Park	Sales Support Agent	Nancy Edwards	Sales Manager
Steve Johnson	Sales Support Agent	Nancy Edwards	Sales Manager
Michael Mitchell	IT Manager	Andrew Adams	General Manager
Robert King	IT Staff	Michael Mitchell	IT Manager
Laura Callahan	IT Staff	Michael Mitchell	IT Manager

QUERY #3: Find big spender by country

I want to analyze which country have the most purchase power by get data from `customers` and `invoices`.

```sql
SELECT
        cus.country,
    SUM(inv.Total) as Amount
FROM customers cus
JOIN invoices inv ON cus.CustomerId = inv.CustomerId
GROUP BY cus.Country
ORDER BY 2 DESC
```

Country	Amount
USA	523.06
Canada	303.96
France	195.1
Brazil	190.1
Germany	156.48
United Kingdom	112.86
Czech Republic	90.24
Portugal	77.24
India	75.26
Chile	46.62

And I add `genres` to get more details that what types of genre they like to buy by country.

```sql
SELECT
        cus.country,
    ge.Name genre_name,
    SUM(inv.Total) as Amount
FROM customers cus
JOIN invoices inv ON cus.CustomerId = inv.CustomerId
JOIN invoice_items itm ON inv.InvoiceId = itm.InvoiceId
```

```
JOIN tracks tr ON tr.TrackId = itm.TrackId
JOIN genres ge ON ge.GenreId = tr.GenreId
GROUP BY 1, 2
ORDER BY 3 DESC
```

Country	genre_name	Amount
USA	Rock	1526.1399999999999
Canada	Rock	989.03
USA	Latin	754.67
Brazil	Rock	691.02
Germany	Rock	577.17
USA	Metal	554.45
France	Rock	525.73
Canada	Latin	510.84
Brazil	Latin	501.93
USA	Alternative & Punk	415.92

QUERY #4: Find total file size in each albums and artist

This one I want to find the total file size in MB by `albums` and `artists`. The simple approach is just join 3 tables: `albums`, `artists` and `tracks` .

But I want to make this query more complicate by using sub-query and CTE (`WITH` clause).

```
WITH trackSize AS (
  SELECT
        trackid,
    albumid,
    ROUND(bytes / 1024.0 / 1024.0, 2) AS MB
FROM tracks
)

SELECT
```

```
        al.Title album_name,
    ar.Name artist_name,
    SUM(MB) AS total_mb
FROM albums al
JOIN artists ar ON al.ArtistId = ar.ArtistId
JOIN trackSize trs ON trs.AlbumId = al.AlbumId
GROUP BY 1, 2
ORDER BY SUM(MB) DESC
```

In sub-query, it is actually `tracks` table but I only select a few fields and convert bytes to megabytes (MB).

And then join table normally.

album_name	artist_name	total_mb
Lost, Season 3	Lost	13272.85
Battlestar Galactica (Classic), Season 1	Battlestar Galactica (Classic)	12276.3
Lost, Season 2	Lost	11773.07
Heroes, Season 1	Heroes	11235.539999999999
Battlestar Galactica, Season 3	Battlestar Galactica	9593.9
LOST, Season 4	Lost	7351.63
The Office, Season 3	The Office	7298.2
The Office, Season 2	The Office	5447.36
Lost, Season 1	Lost	5036.25
The Office, Season 1	The Office	1535.76

QUERY #5: Create file size segment

From QUERY #4, we got total file size by albums and artists. And I want to create group of size segment with window functions

This one I will create 3 segment: Large, medium and small with `NTILE()`.

```
WITH trackSize AS (
  SELECT
        trackid,
```

```sql
    albumid,
    ROUND(bytes / 1024.0 / 1024.0, 2) AS MB
FROM tracks
)

SELECT
        al.Title album_name,
    ar.Name artist_name,
    SUM(MB) AS total_mb,
    NTILE(3) OVER(order BY SUM(MB) DESC) AS segment,
    CASE NTILE(3) OVER(order BY SUM(MB) DESC)
            WHEN 1 THEN 'Large'
        WHEN 2 THEN 'Medium'
        ELSE 'Small'
    END AS segment_name
FROM albums al
JOIN artists ar ON al.ArtistId = ar.ArtistId
JOIN trackSize trs ON trs.AlbumId = al.AlbumId
GROUP BY 1, 2
ORDER BY SUM(MB) DESC
```

album_name	artist_name	total_mb	segment	segment_name
Radio Brasil (O Som da...	O Rappa	168.98	1	Large
The X Factor	Iron Maiden	162.84	1	Large
Dance Of Death	Iron Maiden	155.78	1	Large
Brave New World	Iron Maiden	153.42	1	Large
A-Sides	Soundgarden	148.67	1	Large
Load	Metallica	147.73	1	Large
Pure Cult: The Best Of ...	The Cult	145.48	1	Large
BBC Sessions [Disc 2] [...	Led Zeppelin	144.91	1	Large
Jorge Ben Jor 25 Anos	Jorge Ben	144.51	1	Large
Up An' Atom	Gene Krupa	144.39	1	Large

And we can filter the segment name but the code will be a little complex.

```sql
WITH trackSize AS (
SELECT
        trackid,
    albumid,
    ROUND(bytes / 1024.0 / 1024.0, 2) AS MB
FROM tracks
), segmentTable AS (
SELECT
        al.Title album_name,
    ar.Name artist_name,
    SUM(MB) AS total_mb,
    NTILE(3) OVER(order BY SUM(MB) DESC) AS segment,
    CASE NTILE(3) OVER(order BY SUM(MB) DESC)
            WHEN 1 THEN 'Large'
        WHEN 2 THEN 'Medium'
        ELSE 'Small'
    END AS segment_name
FROM albums al
JOIN artists ar ON al.ArtistId = ar.ArtistId
JOIN trackSize trs ON trs.AlbumId = al.AlbumId
GROUP BY 1, 2
ORDER BY SUM(MB) DESC
)

SELECT *
FROM segmentTable
WHERE segment_name = 'Medium'
```

Here we turn the current query to `WITH` clause. it acts like table and uses with `FROM` clause then I uses `WHERE` to filter segment and that's it!

album_name	artist_name	total_mb	segment	segment_name
Barulhinho Bom	Marisa Monte	108.36	2	Medium
MK III The Final Concer…	Deep Purple	108.3	2	Medium
Revelations	Audioslave	108.16	2	Medium
Volume Dois	Titãs	107.91	2	Medium
Jagged Little Pill	Alanis Morissette	107.9	2	Medium
Cássia Eller - Coleção …	Cássia Eller	107.19	2	Medium
The Best of Ed Motta	Ed Motta	107.16	2	Medium
American Idiot	Green Day	106.83	2	Medium
Unplugged [Live]	Kiss	106.75	2	Medium
Emergency On Planet E…	Jamiroquai	106.43	2	Medium